Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/03/2010
PI Name:	Wise, John Pierce Ph.D.		
Project Title:	Mechanisms of Altered Gravity Potentiation of Metal-Induced Genotoxicity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	John.Wise@usm.maine.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	207-228-8050
Organization Name:	University of Southern Maine		
PI Address 1:	96 Falmouth St		
PI Address 2:	PO Box 9300		
PI Web Page:			
City:	Portland	**State:**	ME
Zip Code:	04103-4864	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	07/01/2009	**End Date:**	06/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received extension to 6/30/2012 per A. Chu/ARC (Ed., 4/29/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AL11G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The long-term objective of this research is to develop a model of the effects of microgravity on the cellular response to genotoxicity. The effects of microgravity on genotoxic response is important because NASA astronauts and personnel in microgravity will be exposed to significant genotoxic agents including chemicals (e.g. metals) and radiation and the potential for microgravity to exacerbate the genotoxic outcome is poorly understood. The short term objective of this research is to initially focus on chromium (Cr) because it is one of the only metals with a well-established DNA repair response to genotoxicity that is also a component of lunar soil. We will investigate the central hypothesis that altered gravity potentiates metal ion- and particle-induced genotoxicity. We will test this hypothesis through two Specific Aims: 1) Determine Altered Gravity Effects on Metal-induced DNA Damage Formation and Repair in Human Cells and 2) Determine if Altered Gravity Effects Are Conserved Across Cell Types and Species. These data when combined with our data generated by our studies of lunar dust in cells at normal gravity can be used to help determine appropriate exposure limits to moon dust. Our data will also be the first to evaluate how altered gravity affects metal-induced DNA repair gene expression and signal transduction as well as effects on metal-induced apoptosis. These data will also indicate which responses are general and which are cell or species specific. Altogether, these data should provide important insight into the mechanisms of altered gravity on metal-induced genotoxicity and suggest possible approaches for potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Aim 1: Our goals for year 1 were as follows: 1) quantify the effect of hyper- and microgravity on metal-induced genotoxicity; 2) measure the effects of altered gravity on metal ion and particle uptake; 3) measure the effects of altered gravity on metal-induced apoptosis; and 4) begin to determine the impact of altered gravity on metal-induced DNA damage repair. Initial progress in year one was delayed because extensive optimization experiments were required to determine several variables before proceeding with the proposed experiments. First a protocol for growing cells on microcarriers had to be developed. We experimented with type of microcarrier, amount of microcarriers, type of cell, seeding density of cells, type of environment to grow the cells and microcarriers (rocking or still), type of trypsin, trypsin time length, and type of rinse to produce the highest number of metaphases per 1500 cells (mitotic index) during a chromosome harvest. These optimization experiments were done on both the human lung fibroblast cells (WHTBF-6) and the human lung epithelial cells. Once optimal growth and reagents were determined we were able to test growth of cells in simulated microgravity using the High Aspect Ratio Vessels (HARV) culture vessels. Based on previous studies (1-3), cells were rotated on the HARV culture vessels at 23 rpm to allow the cells to be in suspension while avoiding excessive centrifugal force. Once all of these parameters were established we were able to begin genotoxicity studies. Three repeats were performed with the ideal parameters. We found that cells grown on microcarriers and rotated at 23 rpm had less damage than cells grown at normal gravity under normal culture conditions. We then needed to confirm that the differences in effects were not due to the fact that cells may grow differently on microcarriers versus normal culture conditions. Experiments were then done in which cells were grown with or without microcarriers at normal gravity. We found there is no significant difference in the amount of chromosome damage seen in cells grown with or without microcarriers at normal gravity. With the wide range of speed settings on the HARV culture vessel rotating base it is possible rotational speed may play a role in the chromosome damage results at 23 rpm in the previous experiments. To test for this we conducted experiments at 8 rpm. This speed is significantly slower than the 23 rpm speed used previously, but is able to grow and maintain cells as demonstrated in previous studies (4-6). According to Hammond, 2001(6) as the rotation approaches zero the spirals of the HARV culture vessels are so large that the particles make one nearly continuous wall impact. At high rotation speeds, wall impacts increase again due to centrifugal forces (7). We have conducted two experiments at 8 rpm thus far, preliminary results suggest that lower rotational speeds have a larger effect. DNA double strand breaks can be measured as a representation of genotoxicity on a finer scale. DNA double strand breaks can serve as an effective measure for other DNA lesions such as adducts, oxidative damage, crosslinks and single strand breaks because the repair of these other lesions can cause the production of DNA double strand breaks or the outcome of their formation may cause replication fork collapse leading to a DNA double strand break. Thus, measuring DNA double strand breaks will address the question of whether altered gravity affects DNA damage on a finer scale either directly as a break or indirectly as another type of lesion. We found that the microcarriers would create significant issues with the proposed comet assay experiments and thus moved to assessing DNA double strand breaks by using a method incorporating g-H2AX detection in flow cytometry. This method allows for immediate detection after exposure regardless of the phase of the cell cycle, while chromosome damage requires cycling cells. In addition, flow cytometry provides valuable information on the effect of microgravity on the cell cycle. The first experiment we have performed thus far experienced issues with the microcarriers. Microcarriers blocked the flow cytometry machine. We are attempting to resolve this issue with the use of cell strainers designed to filter out the microcarriers. Additional experiments are in process. We have also attempted Western Blot assays using our published methods (8) to detect and analyze specific proteins. In the Western Blot procedure cells are scraped rather than trypsinized, however this technique is not possible with cells that grow on microcarriers. In addition, as with the flow cytometry experiments, we are not sure if microcarriers will affect the Western Blot process. This experiment is in progress Aim 2: We have begun to investigate these effects in other species, specifically in bowhead whale and medaka cells. For these cells, we focused on the effect of hypergravity. There was significant time spent in working out the correct parameters and conditions of performing the experiments. Hypergravity was simulated using a centrifuge; however we found that the centrifuge could not generate the temperatures required for proper culture conditions of the cells. The solution to this problem was to run the experiments at room temperature which involved additional testing to ensure that these variables were not influential in the final results. Growth curves were performed in which cells were exposed to room temperature for periods of time equal to those needed for the experiments. We found that this did not affect the overall growth of the cells and thus proceeded with some simple experiments to begin our investigations. We found that the cytotoxic effect of hexavalent chromium is not exacerbated by hypergravity in bowhead whale lung cells or in medaka cells.

	In addition to trouble shooting the basic growth parameters and the correct centrifuge settings to simulate hypergravity, we had to make modifications to our standard protocol for the chromosome damage assay. These experiments are currently being analyzed and the data are pending. References 1. Canova, S.; Fiorasi, F,; Mognato, M. “Modeled Microgravity” affects cell response to ionizing Radiation and Increases Genomic Damage. Radiation Research. 163: 191-199, 2005. 2. Lelkes, P. I.; Galan, D. L.; Hayman, G. T., et al. Stimulated microgravity conditions enhance differentiation of cultured PC12 cells towards the neuroendocrine phenotype. In Vitro Cell. Dev. Biol. 34:316-325; 1998. 3. Sytkowski, A. J.; Davis, K. L. Erythroid cell growth and differentiation in vitro in the simulated microgravity environment of the NASA rotating wall vessel bioreactor. In Vitro Cell. Dev. Biol. 37:79-83, 2001. 4. Maccarrone, M.; Battista, N.; Meloni, M., et al. Creating conditions similar to those that occur during exposure of cells to microgravity induces apoptosis in human lymphocytes by 5-lipoxygenase-mediated mitochondrial uncoupling and cytochrome c release. Journal of Leukocyte Biology. 73: 472-481, 2003. 5. Simons, D. M.; Gardner, E. M.; Lelkes, P.I. Dyamic culture in a rotating-wall vessel bioreactor differentially inhibits murine T-lymphocyte activation by mitogenic stimuli upon return to static conditions in a time-dependent manner. J Appl Physio. 100: 1287-1292, 2006. 6. Liu, T.; Li, X.; Sun, X., et al. Analysis on forces and movement of cultivated particles in a rotating wall vessel bioreactor. Biochemical Engineering Journal. 18: 97-104, 2004. 7. Hammond, T. G.; Hammond, J.M. Optimized suspension culture: the rotating-wall vessel. Am J Physiol Renal Physiol. 281: F12-F25, 2001. 8. Xie H, Wise SS, Holmes AL, Xu B, Wakeman T, Pelsue SC, Singh NP, Wise JP Sr. Carcinogenic lead chromate induces DNA double-strand breaks and activates ATM kinase in human lung cells. Mutation Research, 586:160-172, 2005.
Bibliography Type:	Description: (Last Updated: 08/24/2017)
Abstracts for Journals and Proceedings	McKay J, Wise JP Sr. "Effects of Hypergravity on Chromate-Induced Toxicity in Bowhead Whale Lung Cells and Medaka Fish Fin Cells." University of Southern Maine’s 7th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April, 2010. Proceedings of the University of Southern Maine’s 7th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April, 2010. Abstract # 48, p. 10. , Apr-2010
Abstracts for Journals and Proceedings	Wheeler C, Wise JP Jr, Wise JP Sr. "Effects of Simulated Microgravity on Chromate Toxicity in Human Cells." University of Southern Maine’s 7th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April, 2010. Proceedings of the University of Southern Maine’s 7th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April, 2010. Abstract #49, p. 10. , Apr-2010
Awards	Wise JP Jr. "Maine Space Grant Consortium Fellowship Program. Comparative Toxicology of Human and Bowhead Whale Cells in an Altered Gravity Environment. September 2009." Sep-2009
Awards	McKay J. "Maine Space Grant Consortium Fellowship. Comparative Toxicology of Chromate in Human, Bowhead Whale Lung Cells, and Medaka Fish Fin Cells Under Conditions of Hypergravity. September 2009." Sep-2009